DO THIS FIRST: PINOUT VERIFICATION

WHY YOU ARE READING THIS

Chassis CAN on the 2013 Tesla Model S is on **OBD-II port pin 1 (CH+) and pin 9 (CH-)**, located in the driver's footwell. This is verified from Tinkla's official preAP wiki and matches what gregjhogan, BogGyver, and the openpilot Tesla community use. Some early-build pre-AP cars have these pins UNPOPULATED. You need to check yours visually before plugging anything in.

This procedure: (1) locate and visually inspect the OBD-II port, (2) verify with a multimeter that no pin carries 12V, (3) plug in the SYS TEC and use the sniffer program to confirm Tesla chassis CAN traffic. The angle decode of message 0x370 doubles as a sanity check -- turn the wheel by hand and the displayed angle should follow.

WHAT YOU NEED FOR THIS PROCEDURE

- Digital multimeter with continuity / resistance mode
- SYS TEC USB-CANmodul1 + USB cable
- Laptop with Python and python-can installed (per main manual Steps 1-3)
- Standard OBD-II cable OR pin probes for pins 1 and 9
- **can_sniffer.py** file (in the same folder as this PDF)
- Phone camera (to photograph the OBD-II port for the populated/unpopulated check)

TEST A   VISUAL + MULTIMETER (CAR OFF)

A.1 Locate the OBD-II port: under the dashboard in the **driver's footwell**, near the door, mounted to the underside of the dash. Standard 16-pin trapezoid OBD-II shape. Photograph it with your phone.

A.2 **Inspect pins 1 and 9.** OBD-II pin numbering: top row left-to-right is pins 1-8, bottom row left-to-right is pins 9-16. **Pin 1 is far top-left, pin 9 is far bottom-left.** Compare your photo to the Tinkla wiki reference photos at tinkla.us/index.php/Tesla_Model_S_preAP_OBD2_port -- they show populated vs unpopulated examples side by side.

If pins 1 and 9 are **populated** (have metal terminals visible): you can plug directly into the OBD-II port for chassis CAN access.
If pins 1 and 9 are **empty**: you need a Tinkla Chassis CAN Retrofit Harness (shop.tinkla.us, ~$45) which routes chassis CAN from the diagnostic port behind the MCU cubby down to OBD-II pins 1 and 9. STOP and order one before continuing.

A.3 **Multimeter check** (car OFF, key out). Probe pin 1 to chassis ground and pin 9 to chassis ground. Both should read **0.0V DC**. Tesla OBD-II also has 12V on pin 16 -- DO NOT confuse pin 16 with pin 1.

A.4 Switch multimeter to **resistance / ohms**. Probe between pins 1 and 9. Expected results:

Reading	What it means	Action
~ 60 ohms	Properly terminated CAN bus (two 120 ohm in parallel)	PASS, proceed to Test B
~ 120 ohms	CAN bus with only one terminator, or unterminated stub	Different tap point, or add 120 ohm
Open / >10 kohm	Not a CAN bus, or wires not connected	Try a different pin pair
~ 0 ohms (short)	H and L shorted -- broken bus or wrong pins	STOP. Investigate before continuing
Anything else	Unknown -- not a standard CAN bus	STOP. Try a different connector

Do NOT proceed if Test A fails. Try a different candidate pin pair and run Test A again.

TEST B   PASSIVE LISTEN WITH SNIFFER

Now you connect the SYS TEC and run a program that ONLY listens. It cannot send anything to the bus, so even if your wiring is wrong nothing on the car will be affected.

B.1 Power off your laptop. Connect the SYS TEC DB9 to your candidate pins:

SYS TEC DB9 pin	Wire to:	Notes
Pin 7	OBD-II pin 1 (CH+ chassis CAN high)	required
Pin 2	OBD-II pin 9 (CH- chassis CAN low)	required
Pin 3	OBD-II pin 4 or pin 5 (chassis GND)	recommended

TEST B (continued)

B.2 Power on the laptop. Plug in the SYS TEC. Power the Tesla on (key in, brake pressed for accessory or full Drive Ready). The bus needs to be live.

B.3 Open a command prompt in the folder with the .py files (File Explorer address bar > type cmd > Enter). Run:

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python can_sniffer.py
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B.4 Press Enter when prompted. The script auto-tries 500, 250, and 125 kbps for 4 seconds each. After about 12 seconds total it picks the best baud rate and starts a live capture screen.

WHAT YOU SHOULD SEE

Result	Meaning	Next step
Auto-detect picks 500 kbps with hundreds of frames	You're on a CAN bus running at the right speed.	Continue to B.5
"TESLA CHASSIS CAN CONFIRMED" appears in green	Two or more known Tesla IDs are present. You are on chassis CAN	PASS. Proceed to main manual
"live CAN bus, but no known Tesla IDs"	You found a bus, but it is a different Tesla sub-bus (powertrain, body, etc)	Try a different connector / pin pair
Auto-detect fails at all baud rates	Wires are not on a live CAN bus, or H/L are swapped.	Swap pin 7 and pin 2 wires, retry
Auto-detect picks a baud but error count is high	Bus speed mismatch, termination wrong, or noise.	Check termination (Test A.3), retry

B.5 In the live capture screen, look for these IDs in green:

0x370 EPAS_sysStatus -- the steering rack heartbeat. MUST be present. The script also decodes it: turn the steering wheel by hand and the "measured ang" value should change.
0x101 GTW_epasControl -- the gateway commanding the rack. Should be present at ~10 Hz.
0x129 SteeringAngle -- direct steering wheel sensor. Should be present.

If you see 0x370, 0x101, AND the angle decode tracks the wheel: you are 100% on the right bus. Sign off, move to the main manual.

TEST C ANGLE TRACKING SANITY CHECK

While the sniffer is running, slowly turn the steering wheel left and right by hand. Watch the "measured ang" line at the bottom of the sniffer output. It should:

- Read close to 0 deg with the wheel centered (within +/- 10 deg)
- Increase positive in one direction, negative in the other
- Reach approximately +/- 540 deg at full lock-to-lock (Tesla spec is ~3 turns lock to lock, 1.5 turns each way = 540 deg)

If the angle stays at 0 or jumps around randomly, the EPAS_sysStatus decode is either wrong for your rack firmware OR you are on a different bus that happens to have an ID 0x370 collision. Stop and call Derek.

FINAL DECISION

Test A (multimeter)	Test B (sniffer)	Test C (angle)	Verdict
~60 ohm, no 12V	TESLA CONFIRMED, 0x370 + 0x101 present	tracks wheel	GO. Proceed to main manual.
~60 ohm, no 12V	TESLA CONFIRMED but no 0x370	n/a	WRONG SUB-BUS. Try different pins.
any 12V detected	n/a	n/a	STOP. Wrong pins. Find chassis CAN elsewhere.
120 ohm or open	no traffic	n/a	STOP. Try different pins.

VERIFIED SOURCES FOR THIS PROCEDURE

Pin assignments above are not guesses -- they're cross-referenced from primary sources:

- Tinkla wiki, Tesla Model S preAP OBD2 port: states pins 1 (CH+) and 9 (CH-) carry chassis CAN; provides reference photos of populated vs unpopulated.
- Tinkla wiki, Tesla Giraffe: confirms Pin 1/9 chassis CAN; notes pre-May-31-2013 cars need additional TDC connector for the diagnostic port behind the MCU.
- Tinkla wiki, Panda: confirms openpilot uses OBD-II pin 1/9 as Panda CAN0 = chassis.
- gregjhogan/tesla-pre-ap-epas-patch README: confirms EPAS firmware patching done via panda over chassis CAN bus 0, UDS address 0x730.
- commaai/opendbc tesla_can.dbc: confirms 0x370 EPAS_sysStatus byte/bit positions used by the sniffer's angle decode.

If your build date is BEFORE May 31, 2013: you may need a TDC Connector for additional diagnostic-port access (Tinkla Giraffe wiki). Derek's car is post-May-31-2013 so this does not apply.